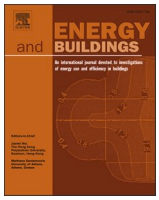
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Comparison of methods for generating ex-ante distribution coefficients for collective self-consumption

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ABSTRACT

Renewable energy communities (REC) based on collective self-consumption (CSC) must allocate the generated electricity to the participant consumers. One possibility for allocation implemented in some countries consists of applying ex-ante distribution coefficients, which might be static or variable. This study assesses 16 modes for generating static and hourly-variable distribution coefficients, using different time resolutions and calculation methods: proportional to consumption (PTC) and numerical optimisation (OPT) to maximise the global self-consumption rate (SCR). The impact on energy and economic Key Performance Indicators (KPIs) is analysed in two case studies in Spain with different configurations of CSC. Case study 1 is composed by a positive energy residential building and a healthcare facility and case study 2 is formed by a school and 159 households. In both case studies, PTC and OPT modes of calculation lead to favourable outcomes for the REC compared to the individual self-consumption scenario. SCR average values for the REC are 86.4% (OPT) and 85.3% (PTC) for case study 1 and 89.2% (OPT) and 88.8% (PTC) for case study 2. OPT modes outperform PTC slightly, with average SCR of the REC for OPT lagging behind the ideal distribution – equivalent to ex-post allocation – in 2 and 3 percentage points for case study 1 and 2 respectively. Within each method, increasing temporal granularity leads to higher values of SCR for the REC. In both case studies 'WDWE + M' is the type of day which leads to highest SCR in all modes. However, increasing temporal granularity entails longer computation times, which reached up to 8.32 min for case study 1 in OPT modes. The improvement obtained with 24 coefficients per day over a constant value for case study 1 is insufficient to offset the increased computational resource. REC economic savings also increased in both case studies in comparison to the base scenario of individual self-consumption, with an increase of 14% and 127% for case study 1 and 2, respectively. Nonetheless this KPI is less affected by the calculation mode chosen and depends on tariff type and consumption patterns. Asymmetries between individual users' results are identified and a compensation for the allocated energy is proposed, which still allows for positive results for the REC. These results provide insights into the most effective criteria for maximising the impact of CSC, raising concern about fair allocation criteria.

1. Introduction

1.1. Motivation

The energy transition, one of the pillars of the ongoing decarbonisation process in the European Union (EU), implies a technological transformation towards renewable technologies and the emergence of new business models and markets. This context presents an opportunity for consumers to assume a more active role in the energy sector, traditionally dominated by large energy companies [1].

In the EU legal framework, energy communities (EC) exist in two forms: 'renewable energy communities' (REC) and 'citizen energy communities' (CEC) [2]. These definitions refer to specific forms or organisations that have in common the purpose of generating social and environmental rather than financial benefits and open and voluntary participation [2,3].

Although the status of the transposition of the EU directives to the national regulation of each member state is heterogeneous [4], many RECs and CECs with different maturity levels are established and currently active. The activities of an EC can include energy generation,

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supply, consumption and sharing, distribution, energy services such as building renovation, electro-mobility, and others, such as information and awareness [2]. Although the most widespread projects involve energy generation [2], many business models apply to ECs given the multiple purposes and motivations for establishing such an entity [5].

Particularly in Spain, generating and sharing renewable electricity represent the most widespread activities within energy community initiatives. The current framework incorporates these activities within the 'collective self-consumption' (CSC) concept, which does not require any legal entity or specific form of organisation. The predominance of this type of project is likely due to a consistent regulatory framework and technology affordability, especially in the case of photovoltaics (PV).

Studies highlight the economic benefits of multiple buildings' shared PV installations compared to individual, especially if the energy is shared between buildings with different types of users [6]. The study shows that integrating diverse load profiles in CSC leads to a greater cost-savings potential [6]. When it comes to CSC without storage, maximising direct self-consumption is the goal for best global results, and this is achieved by matching production and load curves as much as possible. Given the limited control potential of PV production, the option to improve this match is by shifting loads, possibly through consumer behaviour or energy flexibility strategies [7,8].

The current framework for electricity self-consumption in Spain, including CSC, is determined mainly by Royal-Decree 244/2019 [9], which establishes its administrative, technical, and economic conditions. Later modifications from Royal-Decree law 18/2022 include the proximity limits set to the maximum distance of 2,000 m from the producing unit, valid as of September 2025 [10]. Two modalities of self-consumption are distinguished according to the possibility of electricity injection to the grid: with and without surplus electricity. The modality with surplus enables installations up to 100 kW_p nominal power to adhere to the simplified compensation mechanism [9]. This mechanism consists of a net billing system with a compensation value for grid injections. At billing time, the surplus generation exported to the grid is valued at a price determined by the contracted tariff (see Section 2.3 for further details).

Regarding the allocation of the generated electricity, an agreement must be sent to the distribution system operator ex-ante to inform the distribution coefficients of each participant. The electricity generation of each producer installation is measured, given hourly production data is needed for allocation and virtual net metering. Currently, each participant is allowed one coefficient per hour of the year, i.e., 8760 coefficients, [11] and this distribution can be modified every four months. For each hour the sum of the coefficients of all participants must equal one, assuring full allocation of the generated electricity. Other countries' CSC frameworks also allow for static allocation through distribution coefficients, such as France [12] and Portugal [13].

1.2. Research literature

Energy allocation is frequently considered one of the design aspects of ECs, given that it strongly depends on member types and goals, both individual and collective [14]. Ex-ante coefficients can be determined based on the user's contracted power, contribution to the initial investment, or any other criterion agreed upon among all participants [11]. Some modalities applied in real projects are equal distribution, i.e., same coefficient for all participants, and shares of installed power associated with an annual economic fee. Maskova et al. [15] assessed one CEC scenario with constant distribution coefficients in which 30% of the electricity generated is allocated to the building with the PV installation, and the remainder is distributed to the nearby households by shares of 1 kW_p with an annual fee of €100 each. The socio-economic situation of the participants can also be a criterion for allocation, as shown in a CSC model including a tertiary public building and vulnerable residential consumers [16]. Results showed that assigning hourly variable coefficients which allocate the same total yearly electricity to all vulnerable consumers provided the highest savings for this group, although it hindered the potential for savings of the tertiary building [16]. Another study found that for ex-ante distribution coefficients, hourly variable coefficients improved the profitability of CSC projects in comparison to static coefficients [17]. Nonetheless, the extent of the impact depends on specific characteristics of the project (power installed, number of users, and others).

The current regulation in Spain only covers ex-ante static coefficients, but a progressive evolution to ex-post dynamic coefficients is expected, although the legislation mentions the need for deeper analysis and the implementation of a harmonising figure yet to be defined [11]. Dynamic ex-post coefficients show potential to maximise the consumption of locally generated electricity, which implies greater economic savings for the participants [18]. This is particularly relevant in the case of participants with different consumption patterns, whether the variations occur during the week, e.g., offices are closed on weekends, or throughout the year, e.g., a school that is closed in the summer months. A methodology to optimise the allocation of locally generated electricity with ex-post repartition keys aiming to minimise the sum of the electricity bills of all members was tested in [19]. Two strategies were proposed to address possible asymmetries in individual benefits: fixing a minimum self-sufficiency rate for all members and setting initial repartition keys to ensure a minimum revenue per member [19].

The integration of dynamic coefficient strategies, based on consumption forecast, and battery energy storage systems (BESS) was evaluated in [20] and the outcomes showed improved energy and economic Key Performance Indicators (KPIs) for the EC compared to those obtained with ex-ante distribution coefficients. Several methods for calculating dynamic distribution keys were assessed for 8 users with different consumption patterns in [21]. One of the main conclusions is that the allocation strategy leads to better results if the different types of users involved in the EC are considered. Additionally, calculation based on consumption proportion tends to benefit users with the highest consumption, hindering energy savings efforts [21], which must also be fostered within the decarbonisation of the energy sector.

More complex sharing strategies that do not require distribution coefficients include peer-to-peer (P2P) trading, a strategy to overcome the temporal intermittence in renewable energy supply, considering the stochastic characteristics of the energy demand of buildings [22]. In P2P, the sharing criteria also vary, i.e., [23] proposes to maximise social welfare through maximisation of self-consumption of the whole community and P2P trading determined by the willingness-to-pay for local PV generation of each prosumer. Furthermore, allocation through ex-post coefficients and P2P trading expands the possibilities for RECs to apply more complex energy management strategies, such as demand response [24].

Table 1 summarises the reviewed studies, highlighting the country of implementation, the distribution coefficient definition (ex-ante, ex-post, or other), the goal and contribution of the work, and the indicators used to evaluate the results.

1.3. Research gaps

Many methods are described in the literature and proven to reach favourable outcomes for CSC in favour of individual self-consumption. Nonetheless, the current framework of REC and CSC in most EU member states limits their practical applicability. This is the case of Spain and many other member states concerning ex-post distribution coefficients and P2P trading. Considering this, the present study aims to assess allocation based on real conditions of CSC in Spain, which translates to ex-ante distribution coefficients with resolution up to 1 h. From the identified studies, none has explored different calculation modes for ex-ante distribution coefficients, combining temporal resolution with method of calculation: proportional to consumption (PTC) and numerical optimisation (OPT). The different modes defined need to be assessed in terms of relevant energy and economic KPIs and suitability in the

Table 1
Summary of reviewed studies and current study.

Ref.	Country	Ex-ante	Ex-post	Other	Goal / Contribution	Indicators
[15]	Spain	x			Different scenarios evaluated with many producer buildings and associated households, with fix distribution coefficients.	Self-sufficiency rate Total generated electricity Total self-consumed electricity Number of households suppliedSurplus generation.
[16]	Spain	x			Three allocation strategies (equal coefficient, equal savings and equal energy) based on yearly, monthly and hourly temporality for vulnerable residential consumers and non-residential buildings.	Yearly savings (total and per user) Energy allocation per customer category Share of energy self-consumed by the community Share of energy compensated at surplus valueShare of energy not compensated.
[17]	Spain	x			200 scenarios evaluated varying PV capacity, fix or variable distribution coefficients and electricity tariff.	Yearly savingsSimple payback period.
[18]	Austria		x		Assesses electricity flows and costs in a REC with static and dynamic allocation.	Electricity expendituresSelf-sufficiency rate.
[19]	USA		x		Two energy sharing models: a welfare optimisation, and a game theoretical (bi-level) model considering owner of energy assets and consumers.	Total costs Total emissions Total energy generationSurplus energy.
[20]	Spain		x		Dynamic allocation using Day-Ahead energy consumption forecasts generated by Long Short-Term Memory (LSTM) neural networks + BESS integration.	Self-consumption rate Economic savings Payback period Internal rate of return
[21]	Italy		x		Comparison of five methods for dynamic allocation, based on participants' contributions to the community, aiming to achieve fair criteria for energy allocation among the community members, rewarding virtuous behaviour.	Shared energy
[22]	Hong Kong			x	A P2P energy sharing paradigm applied to a system composed of hybrid solar-wind renewable systems, BESS, grid-connected commercial prosumers and flexible energy management.	Self-consumption rate Self-sufficiency rate Imported energy Exported energyEquivalent CO2 emissions.
[23]	Austria			x	Optimisation model maximising the social welfare applied to a community including PV systems and BESS. The PV generation is P2P traded within the community considering each prosumer's individual willingness-to-pay for local PV generation.	Prosumer's annual profitsGHG emissions.
Current study	Spain	x			Two case studies of PV-based CSC in different configurations, applying several methods for calculating distribution coefficients and comparing global and individual energy and economic KPIs.	Self-consumption rate Self-sufficiency rateEconomic savings

context of real RECs.

1.4. Contributions

The study focuses on a comparison between a numerical optimisation-based method (OPT) and a more direct method based on the share of consumption of each participant relative to the total consumption of the REC (PTC) for calculating distribution coefficients of CSC. Within each category, the analysis seeks to provide insights into the appropriate temporal granularity, i.e., how important it is to apply different coefficients depending on the hour of the day, type of day (weekday or weekend), or month to maximise REC's self-consumption. The framework is summarised in Fig. 1. Results are analysed in terms of energy and economic KPIs. Additionally, the study collects data on the computation times required for the optimised method and compares the practical applicability of each assessed mode.

The application to two case studies in Spain intends to provide insight regarding the balance between the simplicity of the application and the results obtained for the REC and for individual participants. The results are expected to provide valuable inputs for making informed

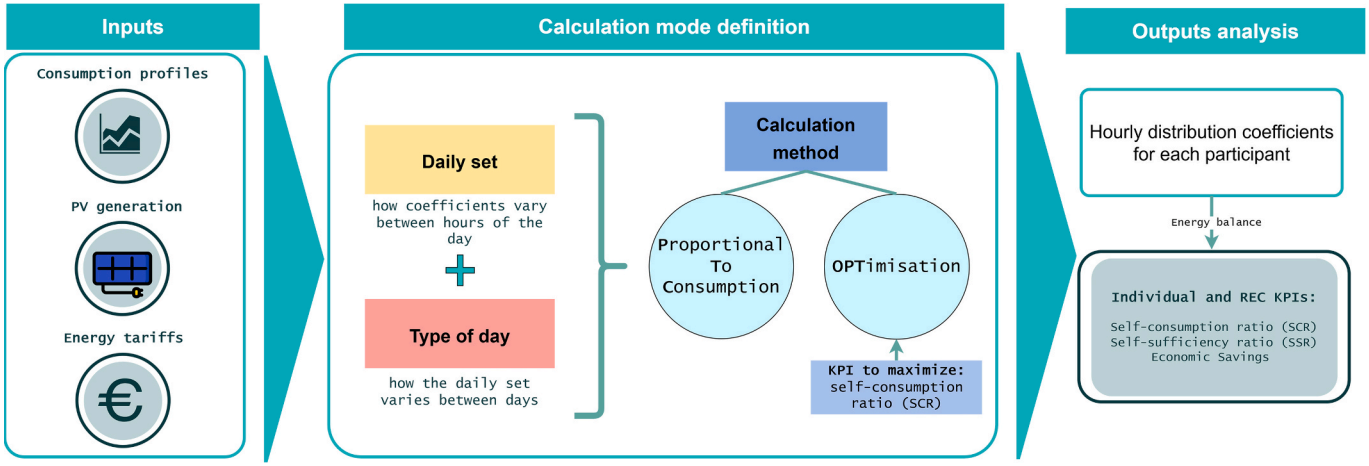


Fig. 1. Proposed framework for distribution coefficients calculation modes assessment.

decisions for energy allocation within CSC, raising concern for a fair distribution of benefits.

The present paper is divided into the following sections: Section 2 describes the materials and methods used in this research; Section 3 analyses and discusses the results, comparing the different methods and case studies; and Section 4 states the main conclusions of the research.

2. Methods

The fundamental concept for REC's energy balance computations is the balance of a single producer–consumer. The total energy consumed, Ec_t, and the total energy generated, Eg_t, are evaluated at every hour t. When a group of n associated consumers share energy from a common power source, the individual energy balances are determined with a set of distribution coefficients β.

$$Eg_{i,t} = \beta_{i,t} Eg_t \quad (1)$$

where $Eg_{i,t}$ is the energy allocated to the consumer i, $\beta_{i,t}$ is the distribution coefficient of consumer i and Eg_t is the total energy generated, at time t. The sum of the set β_i at each time step t must always equal one.

For static coefficients, the distribution coefficient is calculated once for each user, according to the allocation criterion chosen. One example is the proportion of the user's yearly consumption relative to the total annual consumption of all users.

When it comes to determining variable distribution coefficients, the expected consumption patterns and other consumer preferences might be considered. It is possible to calculate it directly based on values of a time series or to implement an optimisation process. In the latter case, possible optimisation objectives include minimising energy exports, total cost, emissions, among others. In this study, several configurations are used to determine the distribution coefficients, applying different time resolutions and calculation methods. The different methods are compared and evaluated on an annual basis by analysing three KPIs [25,26], aligned with relevant KPIs found in the literature (Table 1). The following KPIs are assessed for participants individually and for the REC:

- Self-Consumption Rate (SCR), representing the fraction of total allocated that is directly self-consumed (Eq.2), ranging from 0 (no self-consumption) to 1 (complete self-consumption):

$$SCR = \frac{Total\ self-consumed\ energy}{Total\ allocated\ energy} = \frac{\int min[P_a(t), P_c(t)]\, dt}{\int P_a(t)\, dt} \quad (2)$$

where P_a is allocated power (kW) and P_c is consumed power (kW).

- Self-Sufficiency Rate (SSR) (Eq.3), quantifying the fraction of total energy consumption covered by the energy from self-consumption:

$$SSR = \frac{Total\ self-consumed\ energy}{Total\ consumed\ energy} = \frac{\int min[P_a(t), P_c(t)]\, dt}{\int P_c(t)\, dt} \quad (3)$$

- Yearly savings are calculated by aggregating the monthly bills and comparing them with a reference case (no self-consumption). A detailed description is provided in section 2.3.

The following section describes the different modes for generating the distribution coefficients, followed by the characterisation of the case studies and the economic calculation.

2.1. Calculation modes of ex-ante distribution coefficients

Electricity production and consumption profiles tend to present considerable variability throughout the year, but three prominent patterns can be identified: daily patterns, weekday-weekend variations, and seasonal variations. Several modes are defined to generate the distribution coefficients, combining a daily set with a type of day and assessed with two different calculation methods.

Daily set:

- **Constant (CON)**: single constant value, applied to all hours of the day.
- **24-hour profile (24 h)**: a 24-hourly profile of distribution coefficients, i.e., one coefficient for each hour of the day.

Type of day:

- **Single (S)**: the same set of daily distribution coefficients is applied for the entire year.
- **Weekday-Weekend (WDWE)**: distinct daily sets of distribution coefficients for weekdays and weekends, where consumption patterns may vary.
- **Monthly (M)**: distinct daily sets of distribution coefficients are defined for each month of the year, adapting to seasonal variations in energy consumption and production.
- **WDWE + Monthly (WDWE + M)** combines the previous two, considering different sets of daily distribution coefficients for each month, further differentiated by weekdays and weekends. It accounts for both seasonal and day-type variations.

Calculation method:

- **Optimisation (OPT)**: the coefficients are calculated with a numerical optimisation, aiming to maximise a selected KPI, which in this study is SCR_{REC}. This KPI was chosen as objective function, given that in the current CSC framework maximising self-consumption tends to be the most advantageous situation for REC globally. The constraints of optimization are that all coefficients are positive and the sum of all coefficients for each hour equals 1. Fairness and computation efficiency are not directly included in the optimisation but are analysed especially when comparing calculation modes. The optimisation module is based on *Scipy.Optimize* [27] Python library, more specifically using the Basin-Hopping algorithm which finds the global minimum of the objective function.
- **Proportional to Consumption (PTC)**: the coefficients are set to be proportional to the consumed energy of each consumer relative to the total consumed energy of the REC. The coefficients' values are determined for each consumer as in Eq.4.

$$\beta_{i,t} = \frac{p_{i,t}}{\sum p_{j,t}} \quad (4).$$

where $p_{i,t}$ is the power of consumer i at time interval t. and $p_{j,t}$. is the power of each of the consumers who participate in the REC at time interval t. Note that t will depend on the temporal aggregation of each mode, determined by the daily set and the type of day.

Given the absence of PV production outside the period from 6:00 AM to 8:00 PM, practical sharing is not considered; therefore, no optimisation is carried out for these periods and processing time is reduced. In the case of PTC, coefficients are calculated for every hour of the day, although those outside PV production hours do not have a practical value.

Fig. 2 represents the calculation workflow implemented to obtain the distribution coefficients. The inputs of the module are the consumption and generation profiles of each participant, the participation scheme, i.e. which consumers are eligible to receive energy from specific producers, the energy tariffs and the distribution coefficient mode. The latter is composed of calculation method, daily set, and type of day. For OPT modes, the KPI to maximise is also required, which in this study is always SCR_{REC}. The optimisation module is used only for OPT modes, whereas for PTC modes the calculation of coefficients is embedded in the energy balance function of the Community module. The algorithm performs the energy balance of the REC, determining the distribution coefficient based on the selected mode, and calculates the savings. The

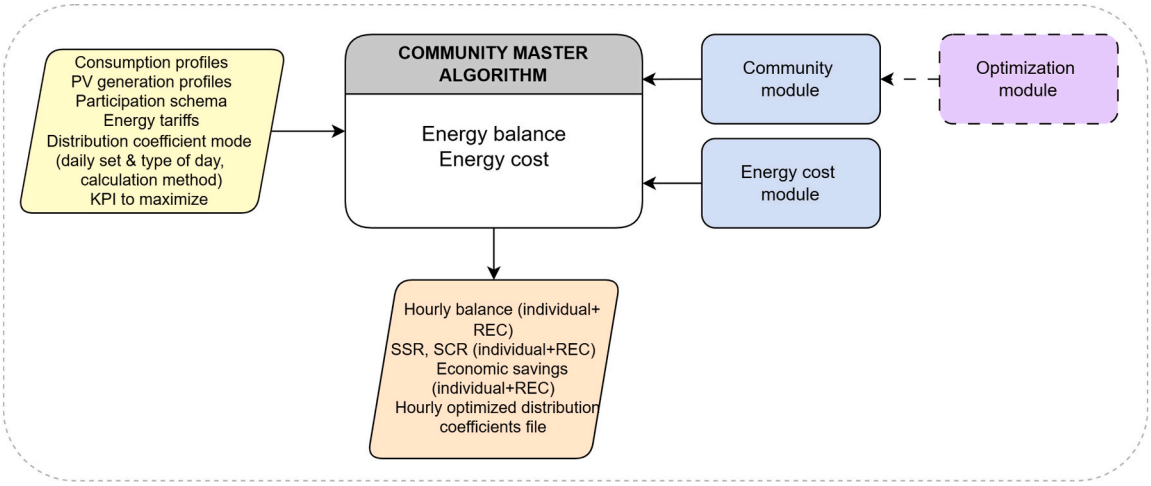


Fig. 2. Distribution coefficient calculation workflow.

outputs are KPIs at individual level (per consumer) and at global level for the REC, and a file with hourly distribution coefficients for each consumer.

The 'ideal distribution' mode is included for comparison, which assumes that all the generated energy is readily available to the community, regardless of which specific user it is allocated to. This would be equivalent to the case of ex-post dynamic allocation, which yields the maximum achievable theoretical value for SCR.

In total, 16 different methods are defined, as shown in Table 2. Case study 1 assesses all modes, whereas for case study 2, the 24 h daily set is not applied due to the reduced improvement of the KPIs observed in case study 1; therefore, only 8 modes are evaluated.

Table 2
Distribution coefficient calculation modes.

Calculation method	Daily set	Type of day	Number of coefficients generated	Abbreviation
IDEAL		–	8760	IDEAL
OPT	CON	Single (S)	1	OPT-CON-S
		WDWE	2	OPT-CON-WDWE
		Monthly (M)	12	OPT-CON-M
		WDWE + M	24	OPT-CON-WDWE + M
	24 h	Single (S)	16 (24)*	OPT-24 h-S
		WDWE	32 (48)*	OPT-24 h −WDWE
		Monthly (M)	192 (288)*	OPT-24 h −M
		WDWE + M	384 (576)*	OPT-24 h −WDWE + M
PTC	CON	Single (S)	1	PTC-CON-S
		WDWE	2	PTC-CON-WDWE
		Monthly (M)	12	PTC-CON-M
		WDWE + M	24	PTC-CON-WDWE + M
	24 h	Single (S)	24	PTC-24 h-S
		WDWE	48	PTC-24 h −WDWE
		Monthly (M)	288	PTC-24 h −M
		WDWE + M	576	PTC-24 h −WDWE + M

* In OPT-24 h mode, optimisation is limited to the 16 distribution coefficients between 6:00 AM and 8:00 PM, corresponding to the hours of PV production. In the table, the actual number calculated is outside parentheses and the theoretical number of coefficients is inside parentheses.

2.2. Description of the case studies

2.2.1. Case study 1: Positive energy residential building and Healthcare facility

Case study 1 is based on a theoretical REC located in Santa Coloma de Gramenet, encompassing a Positive Energy Building for residential use, which is a demonstration-site from the syn.ikia project [28], and a public healthcare facility.

The residential building, shown in Fig. 3, includes the consumption of Common Areas (ComAreas), and the on-site generation of the PV installation (PVres). ComAreas includes the building's centralised energy consumption, including heating and domestic hot water (DHW) centralised systems, mechanical ventilation, lighting of common areas, and elevator consumption.

The healthcare facility comprises two components: the healthcare consumer (HC) and its dedicated photovoltaic system (PVhc). The data employed for the electrical consumption profiles consists of historical hourly consumption from the healthcare building. For the residential building, no real data is available; therefore, consumption profiles are generated through simulations of a detailed building model described in [28,29,30] in TRNSYS18 [31]. The hourly PV generation of both buildings is obtained using TRNSYS18, based on the design included in the executive projects, without any shading effect. The installation of the residential building PVres was sized aiming at maximising the installed power in the available roof area. PVres is composed of 114 solar panels of 355 W_p, with a total installed power equal to 39.1 kW_p. The installation is oriented at 29.7° E, with tilt angle of 30°. The Healthcare installation was limited to one part of the building's roof, with an available area equal to 155 m^2. Therefore, PVhc is composed of 75 PV panels of 400 W_p, with total installed power equal to 30 kW_p.

Fig. 4 summarises the energy consumption of each consumer and the energy generated by each PV installation. Fig. 4a shows that the annual energy generated by PVres is quite close to the total consumption of ComAreas, however, when analysing the annual average daily profile, there is a mismatch between consumption and generation. In a preliminary analysis of individual self-consumption, i.e., no sharing considered, the annual energy balance between PVres and ComAreas results in a SCR of 47.31%. This indicates surplus electricity generation with potential to supply additional consumers, forming a CSC. Additionally, an SSR of 38.68% reflects that less than half of the ComAreas consumption is covered by the on-site PV generation, which is due to the mismatch between load and generation.

On the other hand, the annual consumption of Healthcare is around 4



(a) (b)

Fig. 3. a) residential building b) rooftop with pv installation.
Source: INCASOL

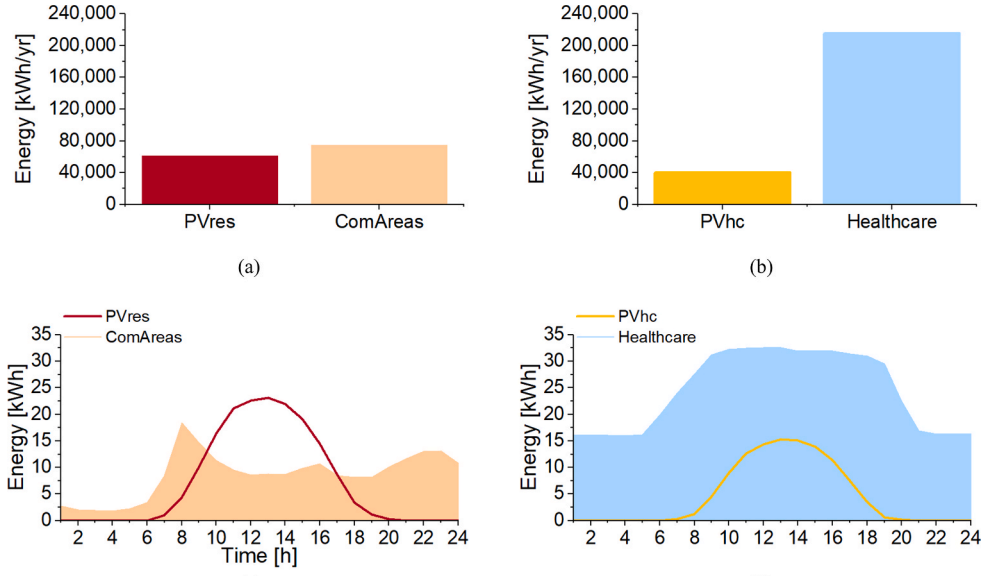


Fig. 4. Case study 1 summary of the energy consumption and the PV generation: a) residential building total annual consumption and generation, b) healthcare total annual consumption and generation, c) residential building average daily profiles of consumption and generation, d) healthcare average daily profiles of consumption and generation.

times higher than the annual energy generated by the PVhc (Fig. 4c), and the average daily profile shows that there is a high consumption during the hours of PV generation (Fig. 4d). In fact, most of its PV generation is consumed on-site, with an SCR of 97.22% yearly. As the energy consumption exceeds the PVhc generation at most times, a much lower SSR is achieved, equal to 18.04%. Thus, Healthcare is a potential consumer for shared self-consumption based on the surplus generation of PVres installation.

As observed in the average daily profiles, buildings have different consumption behaviours, which increases the potential for beneficial synergies in an EC. The following scenarios are considered and represented in Fig. 5:

- *Individual scenario:* no energy sharing is considered, i.e., each producer exclusively supplies energy to its respective assigned consumer. Specifically, PVres provides energy solely to ComAreas, and PVhc supplies energy exclusively to Healthcare.

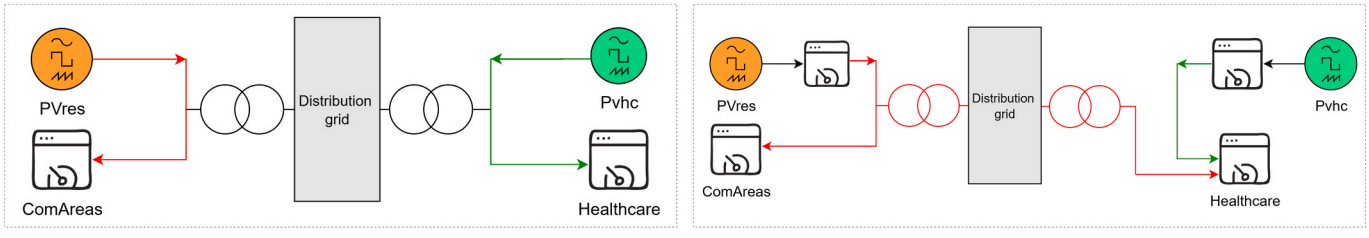


Fig. 5. Case study 1 participation outline: a) individual scenario; b) sharing scenario.

- *Sharing scenario:* considers a CSC with ComAreas and Healthcare as prosumers, and PVres as the shared installation. PVhc continues to supply exclusively to the Healthcare.

2.2.2. Case study 2: School and generic residential units

Case study 2 is based on a theoretical CSC involving the Gabriel Vallseca school, along with neighbouring generic residential households. The school building, part of the ARV project [32], consists of multiple blocks, two stories high, providing rooftop space suitable for PV installations. The building, shown in Fig. 6a, is located in Llevant Innovation District, a neighbourhood from Palma involved in an urban regeneration process.

The analytical framework and key components considered in this case study are depicted in Fig. 7. The Gabriel Vallseca school, named School from now on, functions as a prosumer, given that the PV installation located on its roof is designed to generate electricity for both its own consumption and that of residential participants. The generic residential units, named Households (HHs) from now on, are considered collectively, with aggregated consumption profiles representing typical energy usage within the selected area.

Historical electricity consumption data for the School for the year 2023 forms the basis of the analysis, with total annual consumption of approximately 66.8 MWh. For residential consumption, average hourly profiles from Datadis [33] for Palma's postal code 07,008 is selected, representing a typical residential unit with an annual consumption of approximately 2,795 kWh per household (Fig. 8).

The PV installation parameters are established based on the actual pre-project data provided by EMAYA [34], a municipal utility company from Palma responsible for investing in the PV system. The available rooftop area amounts to 1,315 m^2, accommodating a total of 209 PV panels, resulting in a total installed peak power of 114.95 kWp. The PV system is oriented 41° East and has a tilt angle of 10°. Using tmy file from Palma [35] and Python codes based on the PVGIS methodology [36], the hourly PV production is estimated, without considering shading effects. The annual energy yield from this installation is estimated at 181.6 MWh, with specific energy production reaching approximately 1,579 kWh/kWp annually. The premise for creating a CSC including HHs is to allocate 30% of generated electricity to School and 70% to HHs, initially capped at a maximum export ratio of 15% yearly. Based on this distribution strategy, a total of 159 residential households could be effectively included in the energy sharing arrangement, with each household having a share of approximately 0.5 kWp from the PV installation. It is important to highlight that this is a design criterion for the CSC and that this share will be altered when applying allocation through the distribution coefficients calculated with the different calculation modes.

As seen in Fig. 8, this community configuration highlights a complementary consumption pattern between the school and residential buildings, suggesting significant potential to increase the SCR and generate substantial economic savings for community members. Additionally, during periods of low consumption of the School due to reduced or no activity in summer months, HHs can benefit from a greater share of allocated energy. In consequence, better global results for the REC are expected on yearly basis.

2.3. Economic calculation

To calculate the economic savings for each user, the energy term of the electricity bill is assessed. The power term, which also composes the electricity bill, is considered to remain unchanged, thus it does not imply savings. The savings of each scenario are calculated relative to the bill price before having PV. The tariff considered for consumers with a contracted power exceeding 15 kW (ComAreas and School) is the 3.0TD tariff, based on 6 periods (6P). Households (HHs) savings are assessed with two different tariffs: PVPC (small voluntary price for small consumers), a regulated tariff with hourly-variable prices based on the wholesale market, and 2.0TD tariff, a 3 period tariff (3P). The 3P and 6P tariffs are structured in daily time bands that vary according to month, weekday and location (Mainland Spain for case study 1 and Balearic Islands for case study 2). For 3P and 6P tariffs, the price applicable to each period and the fixed surplus price are market values from March 2025, used to build the hourly tariff time-series for the assessed year. For PVPC, the prices used are hourly historical data from 2024. Fig. 9 illustrates the daily energy prices of all tariffs on a representative day in summer, without taxes. Taxes applied to the monthly bill are 5.113% electricity tax and 21% VAT.

The mechanism of simplified compensation includes a methodology for calculating the value to be charged for the energy term of each billing period, considering surplus electricity.

The calculation of the energy term of each billing period is represented in Fig. 10.

Note that if a negative value is obtained for the energy term, the final value will be 0, given that no monetary value can be obtained by the user. The surplus price applied is constant and equal to 0.065 €/kWh and 0.06€/kWh for tariffs 6P and 3P, respectively. In case of the PVPC tariff, the surplus price varies hourly and the average value of 2024 is 0.062 €/kWh, although the average price in hours of greater PV production is lower, as seen in Fig. 9. From an economic perspective, generally, it is more advantageous to self-consume than to inject surplus electricity into the grid, given the lower price of compensation compared to the market grid prices. No additional costs linked to the generated electricity that might derive from private ownership of the PV installation are

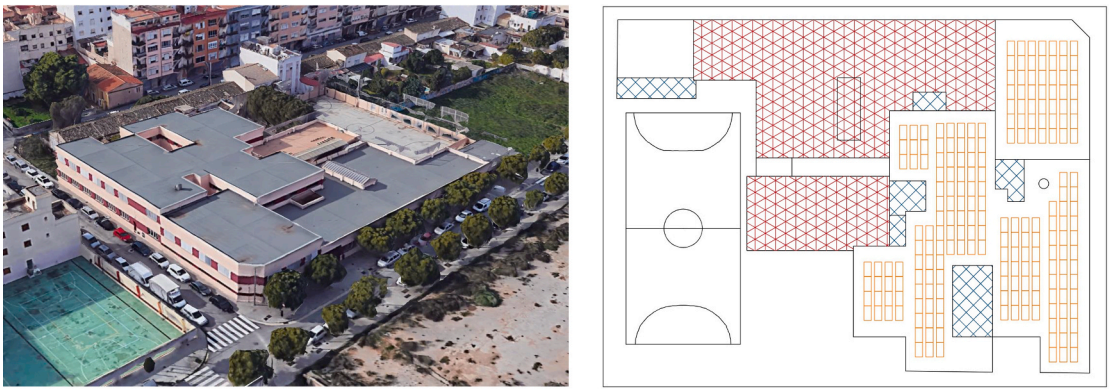


Fig. 6. Case study 2: a) Gabriel Vallseca school building image. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
Source: Google Maps. b) PV installation design, PV panels are represented by orange rectangles, whereas the red zones are reserved for other uses. Source: EMAYA

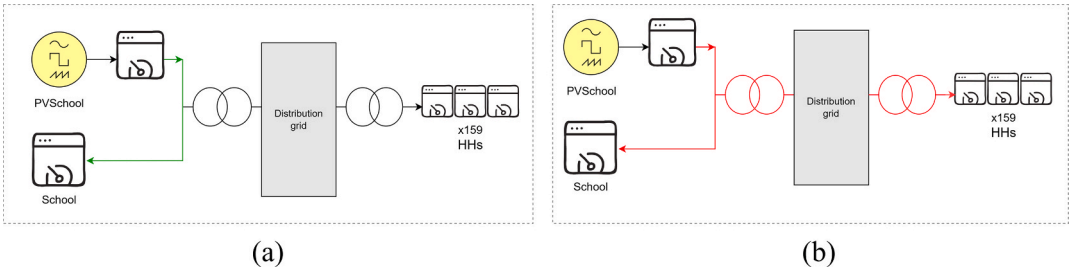


Fig. 7. Case study 2 participation outline a) individual scenario, b) sharing scenario.

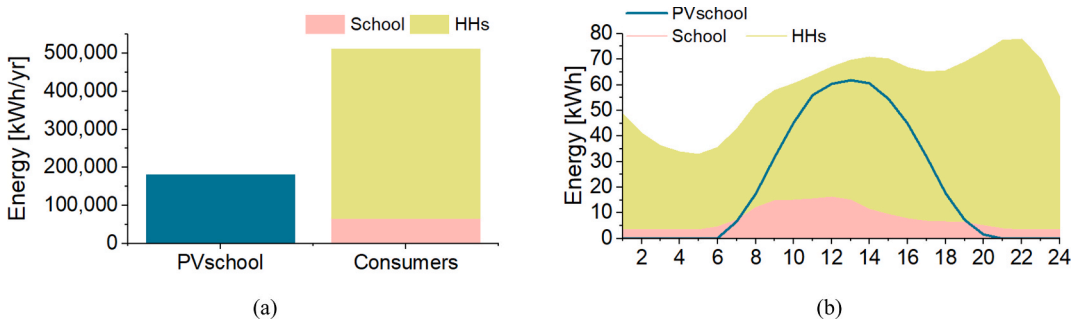


Fig. 8. Case study 2 summary of the energy consumption and the PV generation in each building: a) total annual consumption and generation, b) average daily profiles of consumption and generation.

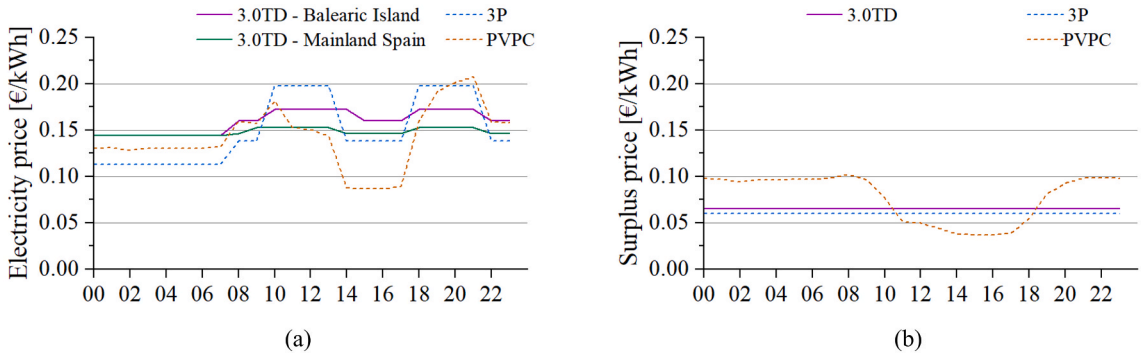


Fig. 9. A) hourly electricity price, b) hourly surplus price for all tariffs on June 7th.

considered, given that the analysis of the installations' investment viability is out of the scope of this study.

3. Results and discussion

In the following sections, the results of the different scenarios and distribution coefficient modes are analysed to evaluate which configuration provides better results in terms of self-consumption and economic savings for each case study.

3.1. Results for case study 1

3.1.1. Case study 1 – energy KPIs

For case study 1, in the individual scenario, KPIs values are $SCR_{Healthcare}$ = 97.2%, $SSR_{Healthcare}$=18.0%, which indicates a mismatch between consumption and production. In turn, $SCR_{ComAreas}$ =47.3% and $SSR_{ComAreas}$ = 38.7%, suggest available energy to share. The maximum achievable SCR_{REC} provided by the ideal distribution is 88.3%. Fig. 11 shows the outcomes of the 16 assessed scenarios for this case study in terms of SCR_{REC} (a) and individual SCR (b). The complete set of KPIs is shown in the Annexes.

Analysing the global results of SCR_{REC} and $Savings_{REC}$, OPT modes generally perform better than PTC, although the differences are modest. The average of SCR_{REC} obtained with OPT modes is 0.9 percentage points above the average of PTC modes. From all modes, OPT-24 h-WDWE + M is the one that yields the highest value for SCR_{REC}, close to those obtained in the ideal scenario. Within OPT modes, the values of SCR_{REC} with a daily set of 24 h are above those obtained from a constant daily set, however, the difference is of only 0.07 percentage points on average. Nonetheless, for PTC method, the trend is opposite, with CON overcoming 24 h in 0.21 percentage points on average for SCR_{REC}. This is due to the limitation of the PTC method, which is programmed to treat each producer as the only power source, overlooking the fact that a consumer may be supplied by multiple sources. This leads to an over-allocation in all scenarios, more pronouncedly in the modes with 24 h daily set.

3.1.2. Case study 1 – Economic savings

The results for REC and individual savings are depicted in Fig. 12. $Savings_{REC}$ equals 17,953€ or 32.2% on average for the 16 sharing modes, which represents an increase of 13.7% in savings compared with the individual scenario. The highest value of $Savings_{REC}$ is obtained with mode OPT-24 h-WDWE + M, the same best-performing mode as for SCR_{REC}.

When analysing individual savings, it is clear that the distribution of the total savings is unequal, while $Savings_{Healthcare}$ responds for 82% of

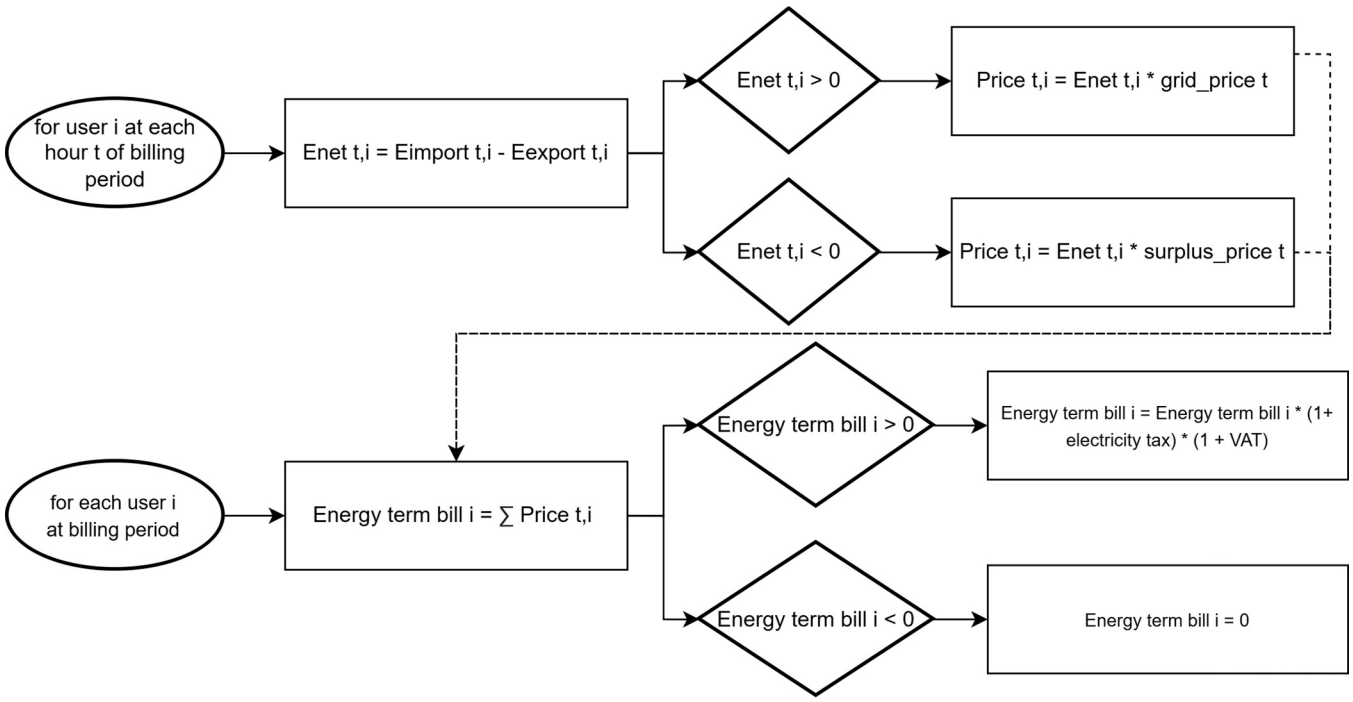


Fig. 10. Flowchart diagram of energy term calculation of each billing period for user i considering simplified mechanism of compensation of surplus electricity.

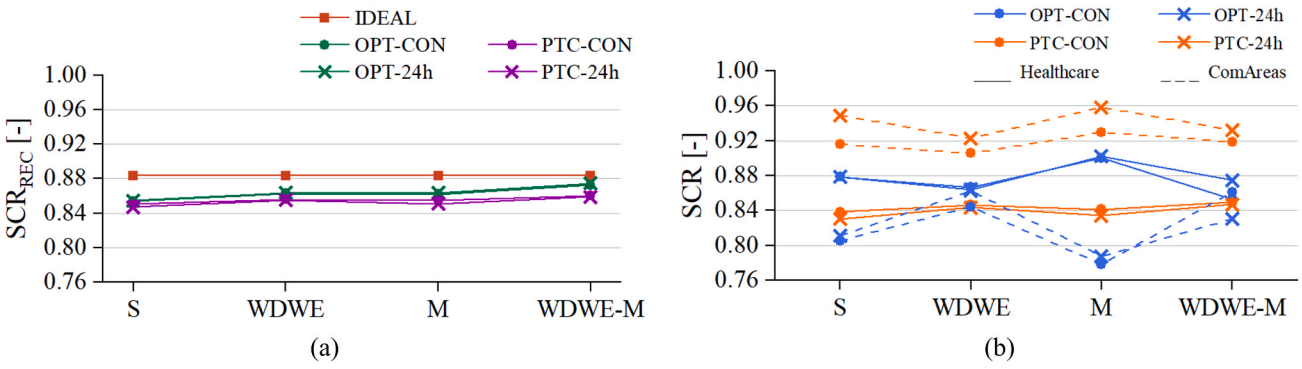


Fig. 11. KPI results for case study 1: a) SCR_{REC}, b) individual participants *SCR* for all modes.

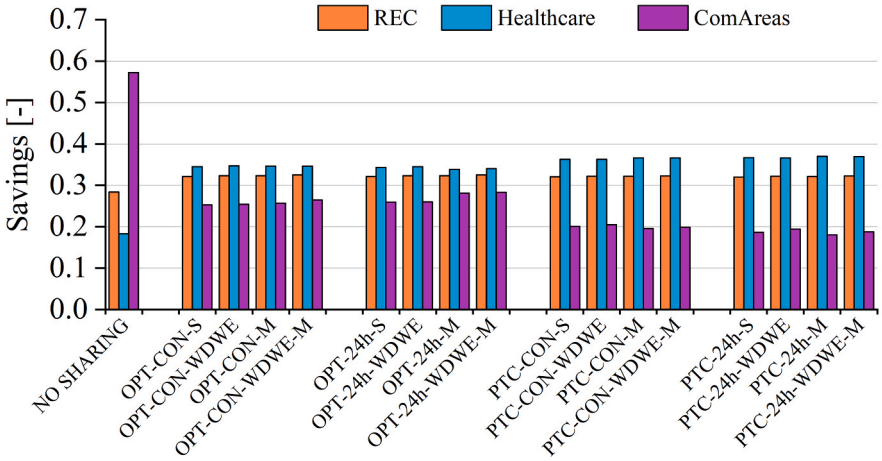


Fig. 12. Case study 1 savings for REC and individual users for all modes.

the total savings, $Savings_{ComAreas}$ contributes with only 18%. Also, it is interesting to notice that compared with the individual scenario $Savings_{Healthcare}$ has a positive increase (93.8% on average) at the expense of $Savings_{ComAreas}$ decreasing its savings (−60% on average). This means comAreas does not benefit economically from sharing part of PV_{res} with the Healthcare. The reduction in the economic savings of ComAreas due to sharing could determine the price of compensation that the Healthcare would pay for the electricity supplied from PV_{res} as a way of neutralising this asymmetry. Concerning the different modes, higher values of $Savings_{Healthcare}$ arise from PTC modes, whereas $Savings_{ComAreas}$ presents higher values in OPT modes. In both cases, the best performing mode is that with greater temporal granularity '-24 h-WDWE + M'.

3.1.3. Case study 1 – computation times of the optimisation

While the PTC modes typically take less than one second of computation to generate a full year's set of distribution coefficients, the process of numerical optimisation of the distribution coefficients is much more time-consuming. Table 3 collects the computation times for each OPT mode in case study 1.

Optimisations are conducted on a dedicated computer equipped with: Microsoft Windows 10 Pro, 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40 GHz, 1382 MHz, 4Cores, 8 Logical Processors, and 16 GB of installed Physical Memory. Fig. 13a provides a general correlation between the quantity of optimised values and the time required to reach an optimal solution.

Fig. 13b represents the computing time for all optimised cases, related to the achieved SCR_{REC}. The modes with a constant daily set (CON) provide similar SCR_{REC} results to the 24 h mode, requiring less computational time. The required processing speed will depend on the type of application, e.g., if the optimisation method is intended for automated processes, lower computation times might be preferred. Fig. 13b may constitute a practical guideline for selecting the most suitable method according to the desired precision and required time.

3.2. Results for case study 2

3.2.1. Case study 2 – energy KPIs

For case study 2, in the individual scenario only the School is self-consuming from the PV installation, so the KPI valuesSCR_{REC} = 24.6% and SSR_{REC} = 10.0% indicate an over-dimensioned installation for individual self-consumption.

Shifting from individual self-consumption to CSC by including residential consumers has a significant effect on SCR_{REC}, with an average of all modes equal to 89.0%, which represents an increase of 64 percentage points. SSR_{REC} reaches an average value of 31.6%, 21.6 percentage points above the individual scenario. Fig. 14 shows the outcomes of the 8 assessed scenarios for this case study in terms of SCR_{REC}(a) and individual *SCR* (b). The complete set of KPIs is detailed in Annexes. Comparing the KPIs obtained with different modes, OPT shows slightly better performance, with average values exceeding those of PTC in 0.43 and 0.15 percentage points for SCR_{REC} and SSR_{REC}, respectively.

Table 3
Case study 1 number and computation times of optimised distribution coefficients.

Mode	Total number of coefficients	Optimisation time (min)
OPT-CON-S	2	0.00
OPT-CON-WDWE	24	0.03
OPT-CON-M	4	0.00
OPT-CON-WDWE + M	48	0.08
OPT-24 h-S	32	0.40
OPT-24 h-WDWE	384	2.90
OPT-24 h-M	64	1.17
OPT-24 h-WDWE + M	768	8.32

The mode which yields the highest value for energy KPIs is OPT-CON-WDWE + M, in fact, SCR_{REC} from this mode lags only 1.9 percentage points behind that from ideal mode. Variation across modes within OPT and PTC methods is not significant, especially for PTC, where the difference of the best and the worst performing mode regarding SCR_{REC} is of only 1.5 percentage points. In both OPT and PTC, '-WDWE' and 'M' type of day show similar outcomes.

3.2.2. Case study 2 – Economic savings

Regarding economic savings for case study 2, Fig. 15 shows savings for all consumers and modes, with PVPC (a) and 3P (b) tariffs. $Savings_{REC}$ equals on average 24,688€ (31.2%) for PVPC and 32,246€ (34.7%) for 3P. The increase obtained in $Savings_{REC}$ for all sharing modes compared to the individual case is highly significant, with an average of 96.8% and 157.1% for PVPC and 3P, respectively. OPT modes lead to greater values of $Savings_{REC}$, although the difference between the average of OPT and PTC is more pronounced when applying PVPC tariff for the HHs, equal to 5.39 percentage points, in contrast to 1.49 percentage points for the 3P tariff. The best performing mode is OPT-CON-WDWE + M, analogous to the result for energy KPIs.

Regarding individual savings, $Savings_{School}$ presents a negative variation of −68.5% on average, which means the school does not get an individual economic benefit from sharing with the residential users. On the other hand, the HHs do not have any savings in the individual scenario, thus they are the most benefited and the responsible for the positive increase of $Savings_{REC}$. The 3P tariff proves to be the one which leads to greatest savings for the residential users and for the REC, both in OPT and PTC modes.

The comparison of results obtained with different modes shows that in the case of $Savings_{School}$ the average of OPT and PTC modes is similar, with values of 31.6% and 29.9%, respectively. Nonetheless, $Savings_{School}$ across OPT modes presents a wider range, varying from 26.2% to 35.5%. OPT-CON-WDWE + M is the mode which presents higher savings and a smaller difference to the individual scenario for School. For HHs, $Savings_{HHs}$ is much less affected by the chosen mode, given that the average of OPT modes and that of PTC are similar for both PVPC and 3P tariff. Similar to the School, $Savings_{HHs}$ in OPT modes present higher dispersion for both tariffs compared to the PTC mode, which achieves similar results for all modes. For PVPC tariff, the mode with highest $Savings_{HHs}$ is OPT-CON-M, whereas for 3P tariff is OPT-CON-S.

3.3. Comparison of case studies

When comparing the two case studies, it is interesting to notice that even though the final REC KPIs are similar for both, the increment obtained compared to the individual scenario is much more pronounced in case study 2. This is likely due to the difference in installation sizing criteria. In case study 1 the installed power is more adjusted to the consumption because the installations are initially planned for individual self-consumption, whereas in case study 2 the installation is dimensioned considering a CSC including nearby households from the start, thus more potential for sharing.

In relation to the different coefficient calculation modes, the best-performing mode for the REC coincided for both case studies, which is constant daily set combined with weekday-weekend type of day and optimised calculation method (OPT-CON-WDWE + M). As expected, this is the mode which can best allocate the energy during weekends and vacation months when one of the participants has a negligible consumption (Healthcare and School). However, the variation observed across modes is not pronounced, especially regarding economic savings, for which maximum deviation, i.e. difference between maximum and minimum values, equals 1.63% for case study 1 and 3.94% (PVPC) and 1.16% (3P) for case study 2. It is important to highlight that the impact of adjusting coefficients aiming at optimum energy allocation has a limited effect because the patterns of the PV production and load curves ultimately constrain the potential for self-consumption. Further work could evaluate the impact of distribution coefficients in more complex REC structures with a larger

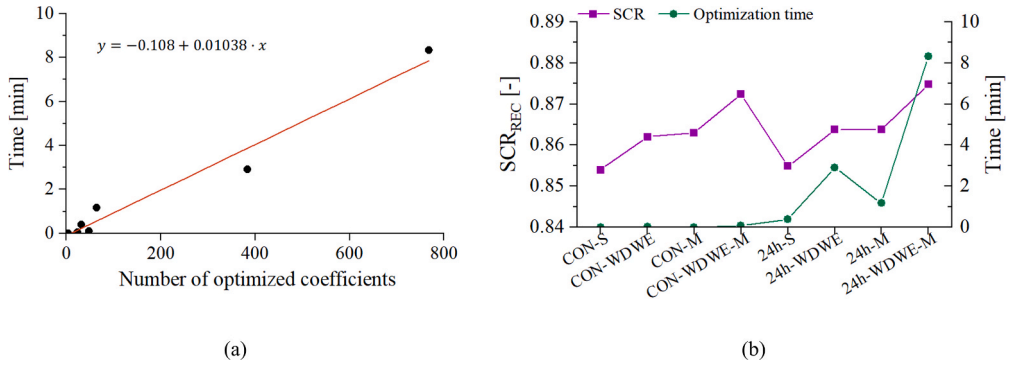


Fig. 13. a) correlation of the computing time and the number of coefficients for each scenario, b) SCR_{REC} and computing time of all the optimised modes.

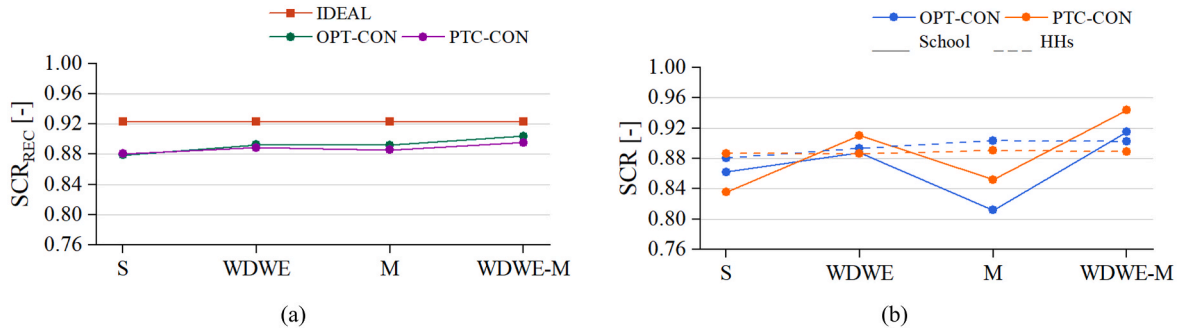


Fig. 14. KPI results for case study 2: a) SCR_{REC}, b) SCR_{School} and SCR_{HHs} for all modes.

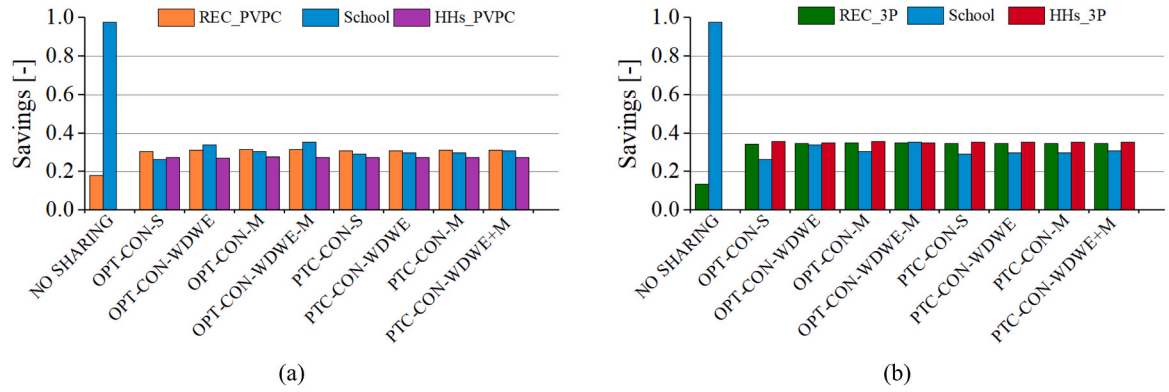


Fig. 15. Savings for case study 2: a) with PVPC tariff for HHs and b) with 3P tariff for HHS. School savings remain unchanged.

number of consumers and greater diversity of profiles. Additionally, the economic benefit is tightly linked to the price of compensation for surplus energy, which is currently quite low. Other strategies with greater impact could be analysed in further work, such as selling surpluses for a higher value through an aggregator or by storing electricity.

In both case studies, important asymmetries among participants are identified in the sharing scenarios, i.e., one consumer benefits from sharing at the expense of another consumer. This asymmetry could be mitigated by allocating costs, through a fixed internal tariff for shared electricity or a yearly participant fee [37]. The cost allocation method can have a significant impact both on the performance of an EC and on participants' perceptions in terms of fairness and transparency [37]. Therefore, the proposed compensation would allow for maintaining global positive results for the REC and promoting a fair allocation of the economic benefit. The goal is that the prosumer recovers the economic losses, calculated in relation to the no-sharing scenario. Considering the best performing mode for case study 1 (OPT-24 h −WDWE + M), the tariff applicable to the Healthcare to compensate the residential building for the shared energy equals 0.112€/kWh, lower than the tariff from any of the 6 periods considered in this study. In case study 2, the annual fee for each household estimated so that the School maintains the savings from the individual scenario, is in the range of 50.2–57.7€. The inclusion of the fee reduces $Savings_{HHs}$ by −42.3% to −37.2%. In practice, compensation schemes may be more complex due to ownership and investment considerations and many studies in the literature address the topics of cost allocation and internal transactions in ECs [38].

4. Conclusions

The constitution of a REC based on CSC with several producers and consumers may require defining distribution coefficients to determine how the generated electricity is allocated to the participants. CSC regulation from Spain and other EU member states includes allocation through ex-ante coefficients, which may be static or variable. The

present work extends the existing literature regarding the application of ex-ante coefficients, calculated with different temporal resolution and calculation method. Two CSC case studies in Spain are analysed to test and compare the different allocation options.

The PTC methods prove to be an effective approach, allocating more energy to consumers with higher demands. The OPT method takes this a step further, achieving higher levels of self-consumption for the REC, although the increase in REC KPIs is not so pronounced. SCR_{REC} average values equal 86.4% (OPT) and 85.3% (PTC) for case study 1 and 89.2% (OPT) and 88.8% (PTC) for case study 2. On average, SCR_{REC} obtained under OPT modes is 2 and 3 percentage points lower than that of the ideal mode – equivalent to ex-post allocation – for case study 1 and 2, respectively. Despite the good results, it is worth noting that the distribution coefficients are defined ex-ante based on historical or simulation data, therefore there is an intrinsic uncertainty in the application of this method. Future work could assess KPIs of ex-ante coefficients applied to a dataset of hourly consumption metered posteriorly.

Regarding temporal granularity, results suggest it is more effective in terms of energy balance and computational time to consider type of day (weekday and weekend) or monthly variations than hourly variations. Case study 1 shows that the improvement obtained by applying 24 coefficients per day does not compensate for the computational effort. Processing time for the most detailed OPT mode reached 8.32 min, which can be considered quite high depending on the application. REC economic savings also increased in both case studies in comparison to the base scenario of individual self-consumption, with an increase of 14% and 127% for case study 1 and 2, respectively. Nonetheless this KPI is less affected by the calculation mode chosen and more dependent on tariff type and consumption patterns.

One important limitation of the case studies is that both deal with only two types of consumption profiles. It is expected that as more consumers with different consumption patterns are involved in the CSC, results obtained with the OPT method stand out more in comparison to PTC..

Collecting hourly consumption data from all CSC participants also adds complexity for the implementation in real cases. Common practice in CSC allocation include modes such as PTC-CON-S and PTC-CON-M, which rely on aggregated annual or monthly data. These modes proved to yield satisfactory results for the REC and may be preferred in some cases for simplicity of application.

Further studies are needed to assess energy and economic balances combined and explore options for allocation which maximise both local energy consumption and economic results for the REC. A more complex optimization could be explored including other KPIs as secondary objectives. Future work could also address more in detail the fairness aspect of allocation, i.e., assuring individual benefits for the REC's members, especially in the case of members in vulnerability.

CRediT authorship contribution statement

Marina Calcagnotto: Writing – original draft, Visualization, Software, Investigation. **Joana Ortiz:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Iuliia Maskova:** Writing – review & editing, Investigation. **Francesc Madrid:** Software. **Jordi Pascual:** Methodology, Conceptualization. **Jaume Salom:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Annexes.

Table 4
Results of selected KPIs for case study 1 obtained with all calculation modes. Values in bold are the maximum for each KPI.

Mode	$SCR_{Healthcare}$ [%]	$SCR_{ComAreas}$ [%]	SCR_{REC} [%]	$SSR_{Healthcare}$ [%]	$SSR_{ComAreas}$ [%]	SSR_{REC} [%]	$Savings_{Healthcare}$ [%]	$Savings_{ComAreas}$ [%]	$Savings_{REC}$ [%]
Individual	97.22	47.31	67.01	18.04	38.68	23.37	18.32	57.22	28.37
IDEAL	–	–	88.35	–	–	30.81	–	–	–
OPT-CON-S	85.52	84.93	85.40	31.99	23.47	29.79	34.53	25.27	32.14
OPT-CON-WDWE	87.87	80.53	86.30	32.51	23.18	30.10	34.71	25.45	32.32
OPT-CON-M	86.67	84.41	86.20	32.26	23.76	30.07	34.61	25.70	32.31
OPT-CON-WDWE + M	89.99	77.83	87.25	32.76	23.72	30.43	34.62	26.46	32.51
OPT-24 h-S	85.33	86.11	85.49	31.75	24.25	29.82	34.33	25.94	32.16
OPT-24 h −WDWE	87.85	81.07	86.37	32.33	23.78	30.13	34.53	26.04	32.33
OPT-24 h −M	86.42	86.25	86.38	31.47	**26.28**	30.13	33.82	28.11	32.35
OPT-24 h −WDWE + M	**90.23**	78.76	**87.48**	32.25	25.50	**30.51**	34.03	**28.32**	**32.55**
PTC-CON-S	83.84	91.57	85.05	33.26	19.33	29.66	36.26	20.11	32.09
PTC-CON-WDWE	84.65	90.57	85.60	33.41	19.64	29.86	36.28	20.47	32.19
PTC-CON-M	84.14	92.98	85.46	33.63	18.85	29.81	36.59	19.53	32.18
PTC-CON-WDWE + M	84.95	91.87	86.01	33.78	19.14	30.00	36.60	19.87	32.28
PTC-24 h-S	83.03	94.93	84.72	33.48	18.25	29.55	36.69	18.65	32.03
PTC-24 h −WDWE	84.33	92.30	85.54	33.64	18.90	29.83	36.61	19.44	32.18
PTC-24 h −M	83.40	**95.84**	85.10	33.86	17.68	29.68	**37.02**	18.03	32.11
PTC-24 h −WDWE + M	84.68	93.24	85.92	**34.02**	18.32	29.97	36.95	18.79	32.26

Table 5
Results of individual and REC KPIs for case study 2 obtained with all calculation modes. Values in bold are the maximum for each KPI.

Mode	SCR_{School} [%]	SCR_{HHs} [%]	SCR_{REC} [%]	SSR_{School} [%]	SSR_{HHs} [%]	SSR_{REC} [%]	$Savings_{School}$ [%]	$Savings_{HHs_{wrc}}$ [%]	$Savings_{REC_PVPC}$ [%]	$Savings_{HHs_3P}$ [%]	$Savings_{REC_3P}$ [%]
Individual	24.56	0	24.56	66.78	0	10.03	97.62	0	18.10	0	13.51
IDEAL	–	–	92.30	–	–	32.78	–	–	–	–	–
OPT-CON-S	86.22	88.08	87.89	24.12	32.29	31.22	26.21	27.57	30.71	**35.84**	34.51
OPT-CON-WDWE	88.75	89.31	89.24	31.59	31.72	31.70	34.04	26.95	31.29	34.87	34.75
OPT-CON-M	81.16	**90.35**	89.20	27.41	**32.33**	31.69	30.61	**27.74**	31.45	35.60	34.91
OPT-CON-WDWE + M	91.48	90.24	**90.40**	**33.53**	31.90	**32.11**	**35.54**	27.25	**31.74**	34.93	**35.02**
PTC-CON-S	83.54	88.64	88.05	26.34	32.02	31.28	29.06	27.32	30.90	35.45	34.57
PTC-CON-WDWE	91.02	88.59	88.87	28.14	32.08	31.57	29.98	27.32	31.02	35.42	34.67
PTC-CON-M	85.19	89.02	88.57	27.22	32.10	31.46	29.69	27.42	31.07	35.47	34.67
PTC-CON-WDWE + M	**94.37**	88.91	89.53	29.48	32.15	31.80	30.81	27.42	31.22	35.44	34.80

Data availability

The authors do not have permission to share data.

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